

Introduction

What is Data Structure?

Data structure is a logical organization of data. It specifies how to store and access the data from memory.

Data may be organized in many different ways; (the logical or mathematical model of a particular organization of data is called a *data structure*.)

Data structure operations

- * **Traversing:** Accessing each element exactly once, is known as traversing. It is also known as visiting.
- * **Inserting:** Adding a new element to data structure.
- * **Deleting:** Removing an element from the data structure.
- * **Searching:** Finding the location of an item in data structure.
- * **Sorting:** Arranging the items in ascending or descending order is known as sorting.
- * **Merging:** Combining the records in two different sorted list into a single sorted list.

Types of Data Structure

- * There are two types of Data Structure: Linear Data Structure and Non Linear Data Structure.
- * **Linear data structure:** A data structure is said to be linear if items are arranged in linear or sequential format.
- * **Non-linear data structure:** Non-linear data structure does not support sequential format.
- * **Linear data structure:**
 - * Array
 - * Stack
 - * Queue
 - * Linked List
- * **Non-linear data structure:**
 - * Tree
 - * Graph

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